Regression	Classification
Multivariable Regression	Logistic Regression
KNN	KNN
SVM (Nonlinear/Gaussian)	SVM (Nonlinear/Gaussian)
Gradient Boost	Adaptive Boost
Random Forest	Random Forest
	Naïve Bayes

- Date and Location: **15:30~18:20, 12/23, RB510**.
- We will do problem lottery in class.
- Regression Problem: Use **at least 4** machine learning models.
- Classification Problem: Use **at least 4** machine learning models.
- You can try stacking in regression or classification if you have lots of time.
- I do not recommend using Neural Network for your final presentation, because it takes lots of time and resources for training and is not explainable!
- Some check lists next page for your reference when building machine learning models

Check List

- Did you use all features or select certain features in your machine learning model?
- Did your data have missing values? If so, how did you treat them?
- Did you have a multicollinearity issue in your linear model?
- Did you check the histogram of the target and features?
- What encoding and scaling did you perform on your data?
- In a classification problem, is your target balanced?
- Binary or Multiclass classification?
- Did you do grid search on hyperparameters for each machine learning model?
- In distance-based models, what distance did you use?
- Did you do cross-validation?
- Did you try something to prevent overfitting?
- Which machine learning models performed better for your problem?
- What metrics did you use for evaluating model performance?
- Did you do SHAP or feature permutation analysis for each model? Were the analysis results consistent across models?
- **If I miss something on this checklist, please let me know. I will make it up!**